

Mouner Wissa

Data Scientist

✉ mw445@njit.edu ☎ 732-791-6041 📍 110 Taylor ave, East Brunswick, NJ 🔗 linkedin.com/in/mouner-wissa-8493a1282
🐙 github.com/Mouner9360 🌐 mouner9360.github.io/portfolio

PROFESSIONAL EXPERIENCE

VR/AR lab research

January 2026 – Present

- **System Engineering:** Developed immersive VR/AR simulations using Unity and C#, architecting real-time data synchronization between head-mounted displays and external servers.
- **Technical Debugging:** Leveraged Git for version control and performed rigorous debugging of C# scripts to maintain system stability across multiple hardware iterations.

Associate Shake shack

February 2024 – August 2025

- Demonstrated flexibility and leadership, transitioning to various roles to accommodate operations needs.
- Ensured operational excellence by managing inventory and supporting kitchen operations with tight deadlines.

Sales Associate Noa Jewelers

October 2021 – October 2022

- Contributed to sales strategy, aligning team efforts to exceed sales targets by 10% for consecutive quarters.
- Provided product knowledge training to new hires, enhancing team effectiveness and ensuring consistent quality.

EDUCATION

Associate Degree computer Science

May 2023

Middlesex community college

Bachelor of Science (B.S.) - Data Science (Computing)

May 2026

New Jersey Institute of Technology (NJIT)

SKILLS

- **AI & Agentic Systems:** Multi-Agent Orchestration (Google ADK), Agentic RAG, Model Context Protocol (MCP), LLM Reasoning (Gemini 1.5/3), Semantic Search, Vector Embeddings, Firecrawl, Claude Code, Claude skills, Prompt Engineering..
- **Machine Learning & Stats:** XGBoost, Random Forest, SHAP (Model Interpretation), Time-Series Forecasting (Lag Analysis), Hyperparameter Tuning (GridSearchCV), KNN Imputation.
- **Programming & Data:** Python (Pandas, NumPy, Scikit-learn, Firecrawl), SQL, Java, C/C++, R, Bash/Linux.
- **Frameworks & Cloud:** Google Cloud Platform (GCP), AWS (EC2/S3), TensorFlow, PyTorch, ZenML/ClearML, Docker.
- **Data Storage & Visualization:** MongoDB Atlas (Vector Search), MySQL, Tableau, Power BI, Matplotlib, Seaborn.
- **Tools:** Git, GitHub, Unity (C#), Model Deployment, ETL Pipelines, Prompt Engineering.

PROJECTS

Autonomous Legal RAG Agent & Infrastructure

3/2026

- **Engineered a High-Performance Ingestion Pipeline:** Automated PDF parsing and Recursive Chunking of U.S. Code, generating 768-dimension vectors via Google text-embedding-004.
- **Optimized Data Throughput by 97%:** Implemented Parallel Processing using ThreadPoolExecutor, leveraging concurrent network connections for bulk embedding generation and MongoDB batch inserts.
- **Architected Agentic Infrastructure:** Developed an MCP-compliant server to interface Gemini 3.1 with MongoDB Atlas, enabling autonomous database administration and vector index management.
- **Developed Custom Retrieval Tools:** Authored a specialized Embedding Tool for real-time prompt transformation, allowing the agent to execute autonomous vector search aggregations on containerized Google Cloud Run infrastructure.

Capstone Project: Lung Cancer Incidence Prediction & Risk Analysis

12/2025

- **Data Engineering & ETL:** Developed a pipeline to merge 20+ annual AQI datasets with CDC health records, processing 14,000+ cleaned entries using KNN Imputation and Standardization.
- **Predictive Modeling:** Optimized XGBoost and Random Forest regressors via 5-fold grid-search cross-validation, achieving an R^2 of 0.96 and a test RMSE of 2.90.
- **Feature Engineering:** Engineered time-series lag features (up to 20 years) and integrated geographical coordinates (Lat/Long), increasing model performance by identifying location as a primary risk predictor.
- **Model Interpretation:** Leveraged SHAP (Beeswarm & Waterfall plots) to quantify the impact of 90th percentile AQI and ozone levels on 5-year cancer incidence rates.

Autonomous Multi-Agent Intelligence System | Google ADK & GCP

2/2026

- **Agentic Orchestration:** Engineered a 3-agent system using Google ADK and Gemini, featuring a Coordinator Agent that autonomously delegates tasks to specialized Research and Summary sub-agents.
- **Automated Web Extraction:** Integrated Firecrawl MCP (Model Context Protocol) to scrape and structure unstructured web data into actionable technical insights.
- **Data Synthesis Pipeline:** Developed a multi-step workflow where a Research Agent analyzes real-time content and a Summary Agent generates comprehensive reports, optimizing research efficiency by 80%.